

Aldwin Von Yu

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SUMMARY

Full-stack software engineer with production experience building business-critical web applications, REST APIs, and database-backed systems in Java, Python, TypeScript, and Node.js. Track record of delivering measurable impact through automation, performance optimization, and clean API architecture — including reducing over 40 hours of weekly manual work through end-to-end system ownership. Experienced with Docker, Linux, PostgreSQL, and CI/CD pipelines. Taught Java and Python at the competitive programming level; strong technical fundamentals across both languages. Fast learner actively building toward Spring Boot, React, and cloud-native development.

EXPERIENCE

Software Engineer / Full-Stack Developer — RoboEdu / VertexStem

Sep 2024 – Present | East York, ON

- Built and maintained a production Django web application supporting class scheduling, homework delivery, student submissions, grading, attendance tracking, and operational communications
- Developed and maintained REST API endpoints for schedule management, attendance, submission, and worksheet workflows; enforced clean separation between business logic and API layers
- Implemented Docker-based isolated execution environments for Java and Python student code, enabling safe and reproducible server-side evaluation of submitted programs
- Automated assignment distribution, grading, feedback, and recurring administrative communications — eliminating over 40 hours of weekly manual work and enabling operational scaling without additional staff
- Designed and optimized PostgreSQL database schemas and queries; reduced long-running queries from ~25 seconds to under 1 second through targeted data-model improvements
- Configured GitHub Actions CI pipeline including Django configuration checks, migration validation, smoke tests, test suite execution, and static asset validation
- Deployed and maintained the application on Linux using Docker and Render; managed environment-driven production settings and secrets configuration
- Implemented auth and account management workflows including login, password reset, token handling, and rate limiting; built SMTP-based email notification and templated communication systems

Lead Programmer — Outlaw Odyssey (Production Indie Project)

Jun 2025 – Present | Remote

- Designed and implemented a turn-based combat system with MVC-style separation between combat logic, event/action queues, animation flow, UI, and scene layout
- Built an internal combat testing tool enabling configurable encounters with reusable player data injection for fast iteration, balancing, and regression testing without launching the full game
- Developed tooling to recalculate character stats from level curves, supporting data-driven gameplay balancing workflows
- Created reusable shader-support tooling for animated sprite behavior, enabling a previously blocked team member to independently author shader-driven animations and increasing animation output by over 100%
- Integrated combat systems into overall game flow and scene transitions; collaborated with designers and artists on iterative feature development

Instructor — Java & Python, Competitive Programming Preparation

Ongoing | Toronto, ON

- Taught data structures and algorithms in Java and Python to students preparing for ACSL (American Computer Science League) and CCC (Canadian Computing Competition)
- Designed and delivered lessons on core CS fundamentals including sorting, searching, recursion, and graph traversal using Java and Python implementations

EDUCATION

Bachelor of Economics — University of Waterloo

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS, GDScript **Back-End & Frameworks:** Django, Node.js, REST APIs, server-side architecture **Databases:** PostgreSQL, SQLite **DevOps & Infrastructure:** Docker, Linux, Git, GitHub Actions CI/CD **Software Engineering:** OOP, data structures & algorithms, system design, unit testing, full SDLC **Learning Focus:** Spring Boot, React, OpenShift, cloud-native development, LLM / Generative AI concepts **Methodologies:** Agile delivery, stakeholder-driven requirements, technical documentation

PROJECTS

Serverless REST API – Email Summarization Service

Tech: TypeScript, Node.js, Fermyon Cloud (Spin), OAuth 2.0, Gmail API

- Designed and implemented a serverless REST API end-to-end using TypeScript and Node.js
- Integrated webhook-based requests from a Telegram bot to trigger multi-system API workflows
- Implemented OAuth 2.0 authentication to securely access the Gmail API; stored tokens using Fermyon key-value storage without persisting user data
- Implemented cron-triggered scheduled execution and async request handling to minimize execution time and serverless resource usage
- Applied structured error handling and logging across API boundaries; built around AI-adjacent email summarization workflows demonstrating practical familiarity with LLM-driven automation patterns